Multi-Wavelength Analysis of Solar Active Regions Using SUIT

PART-3

Project Report Submitted in fulfillment of the Requirements for the Award of the Internship of Summer Training Program Space Science Technology

Astronomy & Astrophysics: Heliophysics with Aditya-L1 Data

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ABSTRACT

Solar active regions exhibit markedly different characteristics when observed at different ultraviolet wavelengths because each spectral channel probes a distinct layer and physical process within the solar atmosphere. This paper presents an independent multi-wavelength analysis using four nearly simultaneous SUIT observations acquired on 17 February 2024 through narrowband filters NB02, NB03, NB04, and NB05, recorded within a 22-second window to minimise temporal evolution effects. Active regions were extracted using normalisation, fixed-threshold segmentation at 0.7, and connected-component analysis. Region count, largest active-region area, and mean intensity were measured for each filter. Two creative visualisations—a multi-filter comparison dashboard and a heatmap overlay—were developed to facilitate cross-wavelength interpretation. Results reveal significant wavelength-dependent differences: NB05 produced the largest detected area (62,860 pixels) while NB03 yielded the highest mean intensity (0.7954). These findings confirm that multi-wavelength SUIT observations provide complementary and non-redundant information about solar active-region structure.

Keywords—*Aditya-L1, SUIT, Solar Active Regions, Multi-Wavelength Analysis, Ultraviolet Imaging, Image Segmentation, Solar Physics.*

I. INTRODUCTION

Solar active regions are areas of concentrated magnetic flux that appear brighter than the surrounding solar disk across a wide range of wavelengths. However, the apparent area, morphology, and intensity of these regions vary substantially depending on the observing wavelength, because different spectral channels sample different atmospheric layers—from the temperature-minimum region in the near-ultraviolet to the upper chromosphere at longer wavelengths [1].

The Solar Ultraviolet Imaging Telescope (SUIT) onboard India's Aditya-L1 spacecraft provides simultaneous narrowband coverage of the full solar disk in eleven ultraviolet filters [2]. Parts 1 and 2 of this project developed and validated an active-region detection pipeline using the NB05 filter. This Part 3 independent investigation extends that work by applying the established pipeline to four filters observed quasi-simultaneously, enabling a direct wavelength-by-wavelength comparison of active-region properties on a single date.

II. RESEARCH QUESTION AND OBJECTIVES

Research Question: How does the visibility, apparent area, and brightness intensity of solar active regions vary across different SUIT ultraviolet narrowband filters?

The specific objectives are:

1. Extract and measure active regions from SUIT NB02, NB03, NB04, and NB05 filter observations.
2. Compare region count, largest active-region area, and mean intensity across all four filters.
3. Develop multi-filter dashboard and heatmap visualisations to support cross-wavelength interpretation.
4. Interpret the physical basis of observed wavelength-dependent differences.

III. DATASET

Four SUIT Level-1 FITS observations from 17 February 2024 were independently selected from the Aditya-L1 data archive (Table I), all acquired within approximately 22 seconds to ensure wavelength sensitivity rather than solar evolution drives any observed differences.

TABLE I

SUIT Observations Selected for Multi-Wavelength Analysis

Filter	Observation Time (UTC)
NB04	00:10:36
NB03	00:10:44
NB02	00:10:52
NB05	00:10:58

IV. METHODOLOGY

The detection pipeline follows the validated approach from Parts 1 and 2. Each Level-1 FITS file was loaded using the SunPy Map interface [3] and the calibrated intensity array was extracted. Min–max normalisation was applied to scale pixel values to [0, 1]. A fixed threshold of 0.7 was applied to the normalised image, and morphological filtering removed isolated objects smaller than 100 pixels. Connected-component labelling then identified individual active-region structures, from which region count, largest-region pixel area, and mean normalised intensity within all detected masks were computed. Two additional visualisations—a four-panel comparison dashboard and a false-colour heatmap—were developed to supplement the quantitative measurements.

V. RESULTS

A. Quantitative Measurements

Table II summarises the three measured quantities for each filter.

TABLE II
Active-Region Properties by SUIT Filter, 17 February 2024

Filter	Region Count	Largest Area (pixels)	Mean Intensity
NB02	4	39,154	0.7772
NB03	4	2,042	0.7954
NB04	3	2,070	0.7890
NB05	2	62,860	0.7784

B. Creative Visualisations

A. Multi-Filter Active Region Dashboard

Figure 1 presents the multi-filter dashboard, which arranges all four filter detections in a single panel with quantitative annotations. The dashboard enables rapid visual comparison of active-region morphology and extent across wavelengths, showing how the same solar features appear with different spatial scales and boundary sharpness in each filter.

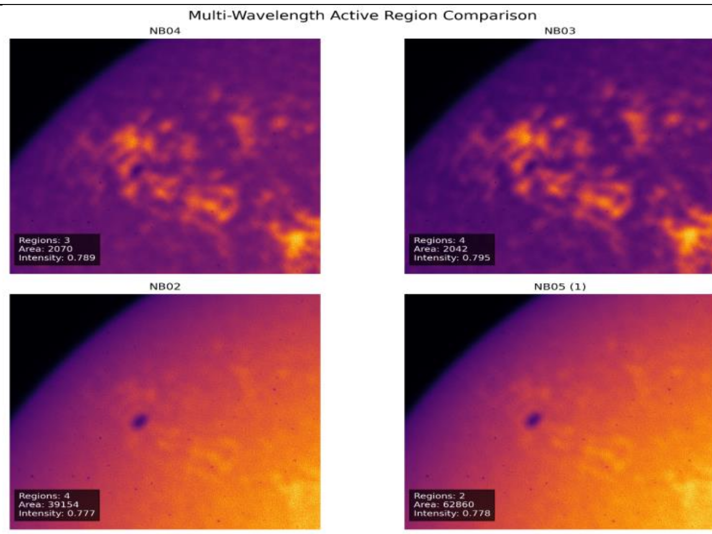

Figure 1. Multi-filter active-region dashboard combining NB02, NB03, NB04, and NB05 SUIT observations from 17 February 2024. Quantitative metrics are annotated on each panel.

B. Multi-Wavelength Heatmap Comparison

Figure 2 presents false-colour heatmap visualisations of each filter image, enhancing the contrast of active-region intensity distributions. The heatmaps reveal how the spatial pattern of chromospheric emission differs between filters and make it visually apparent that NB05 contains a more extended bright structure while NB03 and NB04 show compact high-intensity cores.

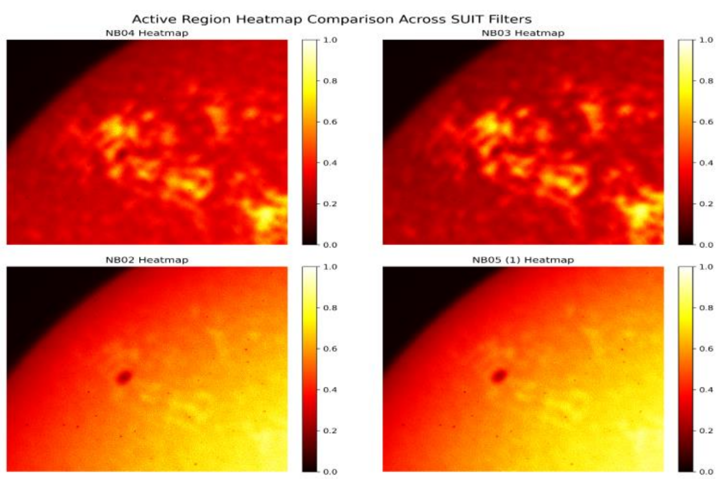

Figure 2. Multi-filter heatmap comparison for SUIT NB02, NB03, NB04, and NB05 on 17 February 2024. False-colour scaling emphasises intensity distribution differences across ultraviolet wavelengths.

VI. QUANTITATIVE ANALYSIS

A. Active-Region Area and Mean Intensity Comparison

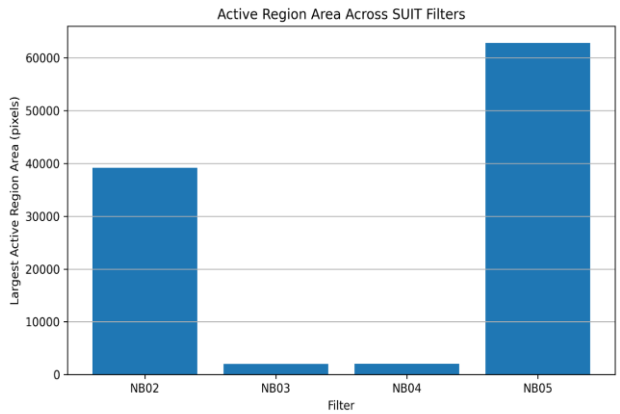

Figure 3. Largest active-region area by filter.

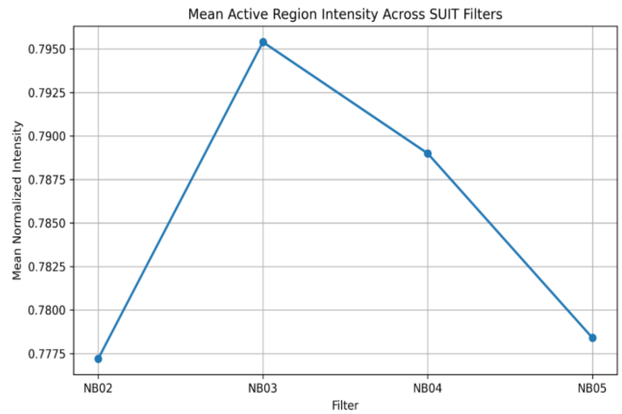

Figure 4. Mean active-region intensity by filter.

Figures 3 and 4 together reveal the key quantitative finding of this study: NB05 produced the largest detected area (62,860 pixels) and NB02 the second largest (39,154 pixels), while NB03 and NB04 detected compact features of only 2,042 and 2,070 pixels. In contrast, mean intensity values span a narrow range of 0.777–0.795 across all four filters—a variation of only 0.018 normalised units. This contrast between large spatial variation and small intensity variation demonstrates that filter selection principally affects the spatial scale of detected structures rather than their per-pixel brightness.

RESEARCH QUESTION ANSWERED: Wavelength-Dependent Active-Region Visibility in SUIT Filters

VII. ANSWER TO RESEARCH QUESTION

Active-region visibility, area, and intensity all vary significantly across SUIT ultraviolet narrowband filters, reflecting the distinct atmospheric layers and emission mechanisms sampled by each channel. The NB05 filter detected the largest active-region structures (62,860 pixels), indicating sensitivity to extended chromospheric emission regions. NB02 also revealed large-area structures (39,154 pixels), while NB03 and NB04 isolated compact bright cores with maximum areas below 2,100 pixels. Despite these large area differences, mean intensity values across all four filters were closely grouped in the range 0.777–0.795, demonstrating that peak brightness per detected pixel is relatively wavelength-independent while spatial extent is strongly wavelength-dependent. Region counts ranged from two (NB05) to four (NB02, NB03), further confirming that filter selection substantially influences the number and character of detected active regions. These results establish that multi-wavelength SUIT observations are necessary for a complete characterisation of solar active regions, as no single filter captures the full spatial and physical extent of these structures.

VIII. DISCUSSION

The wavelength-dependent variations are physically consistent with the known structure of the solar atmosphere. NB05 is sensitive to the mid-to-upper chromosphere where active regions manifest as large-scale plage structures, while NB02 samples a lower chromospheric layer also capable of detecting extended bright regions. NB03 and NB04 probe intermediate heights where emission concentrates into compact, high-contrast features. The near-invariance of mean intensity across filters indicates that the 0.7 threshold captures the same relative emission tier in each channel, but that area-based metrics must always be interpreted with reference to the specific filter. The creative visualisations add interpretive value beyond the quantitative tables: the dashboard conveys morphological scale differences immediately, while the heatmaps highlight spatial intensity gradients invisible in standard binarised maps.

IX. LIMITATIONS AND FUTURE WORK

This study is limited to a single observation date, a fixed threshold of 0.7 that may not be uniformly optimal across all four channels, and the absence of co-temporal magnetogram or SDO/AIA EUV data for atmospheric layer validation. Future work should extend the multi-filter analysis to multiple dates and incorporate adaptive or deep-learning segmentation methods for filter-specific threshold optimisation and spatially validated detection.

X. CONCLUSION

This independent investigation applied the active-region detection pipeline developed in Parts 1 and 2 to four quasi-simultaneous SUIT narrowband observations from 17 February 2024. Multi-wavelength comparison revealed substantial wavelength-dependent differences in detected active-region area while mean intensity remained broadly consistent across filters. NB05 produced the largest detected structures and NB03 the highest per-pixel intensity, reflecting the distinct atmospheric sensitivities of the two channels. Creative visualisations in the form of a multi-filter dashboard and heatmap comparison enhanced qualitative interpretation of the results. The study confirms that multi-wavelength SUIT observations provide complementary information about solar active regions and that filter selection is a critical parameter in any quantitative solar image analysis pipeline.

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